

Graduate Aptitude Test in Engineering

Notations :

- Options shown in green color and with ✓ icon are correct.
- Options shown in red color and with ✗ icon are incorrect.

Question Paper Name: EE: ELECTRICAL ENGINEERING 7th Feb Shift1
Number of Questions: 65
Total Marks: 100.0

Wrong answer for MCQ will result in negative marks, (-1/3) for 1 mark Questions and (-2/3) for 2 marks Questions.

General Aptitude

Number of Questions: 10
Section Marks: 15.0

Q.1 to Q.5 carry 1 mark each & Q.6 to Q.10 carry 2 marks each.

Question Number : 1 Question Type : MCQ

Didn't you buy _____ when you went shopping?

- (A) any paper (B) much paper (C) no paper (D) a few paper

Options :

- ✓ A
- ✗ B
- ✗ C
- ✗ D

Question Number : 2 Question Type : MCQ

Which of the following options is the closest in meaning to the sentence below?

She enjoyed herself immensely at the party.

- (A) She had a terrible time at the party
(B) She had a horrible time at the party
(C) She had a terrific time at the party
(D) She had a terrifying time at the party

Options :

- ✗ A
- ✗ B
- ✓ C
- ✗ D

Question Number : 3 Question Type : MCQ

Which one of the following combinations is incorrect?

- (A) Acquiescence - Submission
- (B) Wheedle - Roundabout
- (C) Flippancy - Lightness
- (D) Profligate - Extravagant

Options :

- 1. ✗ A
- 2. ✓ B
- 3. ✗ C
- 4. ✗ D

Question Number : 4 Question Type : MCQ

Based on the given statements, select the most appropriate option to solve the given question.

If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building?

Statements:

- (I) Each step is $\frac{3}{4}$ foot high.
- (II) Each step is 1 foot wide.

- (A) Statement I alone is sufficient, but statement II alone is not sufficient.
- (B) Statement II alone is sufficient, but statement I alone is not sufficient.
- (C) Both statements together are sufficient, but neither statement alone is sufficient.
- (D) Statement I and II together are not sufficient.

Options :

- 1. ✓ A
- 2. ✗ B
- 3. ✗ C
- 4. ✗ D

Question Number : 5 Question Type : MCQ

Given Set A = {2, 3, 4, 5} and Set B = {11, 12, 13, 14, 15}, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

- (A) 0.20
- (B) 0.25
- (C) 0.30
- (D) 0.33

Options :

- 1. ✓ A
- 2. ✗ B
- 3. ✗ C
- 4. ✗ D

Question Number : 6 Question Type : MCQ

Select the alternative meaning of the underlined part of the sentence.

The chain snatchers took to their heels when the police party arrived.

- (A) took shelter in a thick jungle
- (B) open indiscriminate fire
- (C) took to flight
- (D) unconditionally surrendered

Options :

- 1. ✖ A
- 2. ✖ B
- 3. ✔ C
- 4. ✖ D

Question Number : 7 Question Type : MCQ

The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option.

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

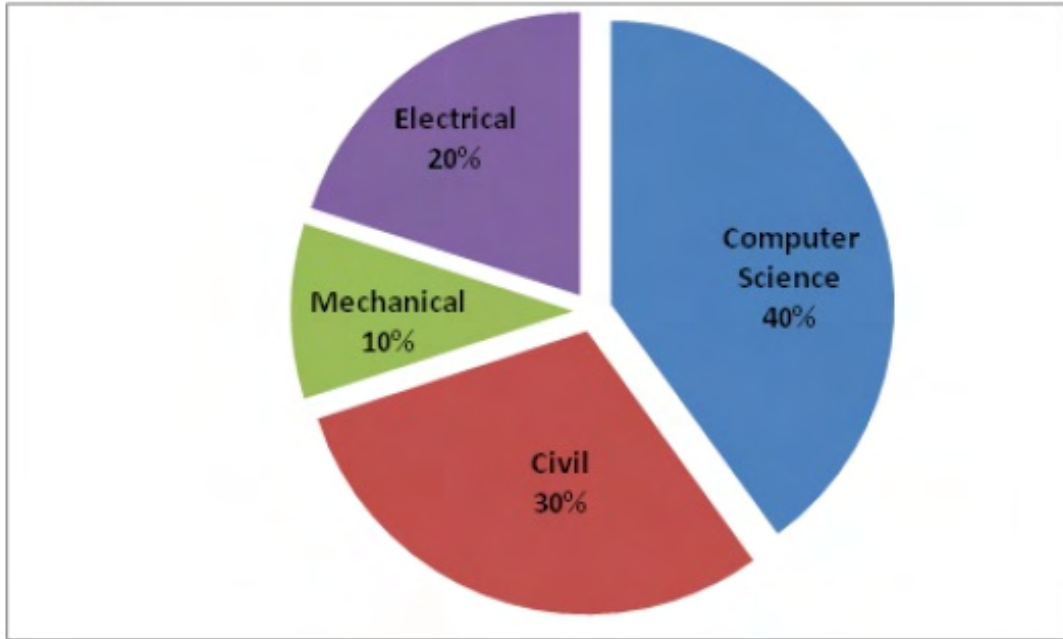
- (I) The water supply authority should impose a partial cut in supply to tackle the situation.
 - (II) The government should appeal to all the residents through mass media for minimal use of water.
 - (III) The government should ban the water supply in lower areas.
-
- (A) Statements I and II follow.
 - (B) Statements I and III follow.
 - (C) Statements II and III follow.
 - (D) All statements follow.

Options :

- 1. ✔ A
- 2. ✖ B
- 3. ✖ C
- 4. ✖ D

Question Number : 8 Question Type : NAT

The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5:4. There are 40 males in Electrical Engineering. What is the difference between the numbers of female students in the Civil department and the female students in the Mechanical department?



Correct Answer :

32

Question Number : 9 Question Type : MCQ

The probabilities that a student passes in Mathematics, Physics and Chemistry are m , p , and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m , p , c :

- (I) $p + m + c = 27/20$
- (II) $p + m + c = 13/20$
- (III) $(p) \times (m) \times (c) = 1/10$

- (A) Only relation I is true.
- (B) Only relation II is true.
- (C) Relations II and III are true.
- (D) Relations I and III are true.

Options :

- 1. ✗ A
- 2. ✗ B
- 3. ✗ C
- 4. ✓ D

Question Number : 10 Question Type : MCQ

The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

- (A) 2.290 (B) 2.970 (C) 6.795 (D) 8.795

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Electrical Engineering

Number of Questions: 55
Section Marks: 85.0

Q.11 to Q.35 carry 1 mark each & Q.36 to Q.65 carry 2 marks each.

Question Number : 11 Question Type : NAT

A random variable X has probability density function $f(x)$ as given below:

$$f(x) = \begin{cases} a + bx & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

If the expected value $E[X] = 2/3$, then $Pr[X < 0.5]$ is _____.

Correct Answer:

0.25

Question Number : 12 Question Type : MCQ

If a continuous function $f(x)$ does not have a root in the interval $[a, b]$, then which one of the following statements is TRUE?

- (A) $f(a) \cdot f(b) = 0$
- (B) $f(a) \cdot f(b) < 0$
- (C) $f(a) \cdot f(b) > 0$
- (D) $f(a)/f(b) \leq 0$

Options :

- 1. ✖ A
- 2. ✖ B
- 3. ✔ C
- 4. ✖ D

Question Number : 13 Question Type : NAT

If the sum of the diagonal elements of a 2×2 matrix is -6 , then the maximum possible value of determinant of the matrix is _____.

Correct Answer :

9

Question Number : 14 Question Type : MCQ

Consider a function $\vec{f} = \frac{1}{r^2} \hat{r}$, where r is the distance from the origin and $\hat{r}$ is the unit vector in the radial direction. The divergence of this function over a sphere of radius R , which includes the origin, is

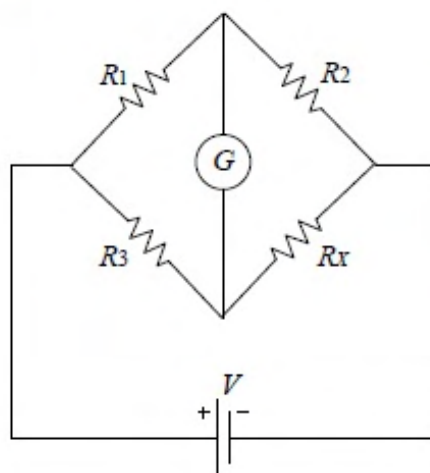
- (A) 0 (B) 2π (C) 4π (D) $R\pi$

Options :

- 1. ✖ A
- 2. ✖ B
- 3. ✔ C
- 4. ✖ D

Question Number : 15 Question Type : MCQ

When the Wheatstone bridge shown in the figure is used to find the value of resistor R_X , the galvanometer G indicates zero current when $R_1 = 50\ \Omega$, $R_2 = 65\ \Omega$ and $R_3 = 100\ \Omega$. If R_3 is known with $\pm 5\%$ tolerance on its nominal value of $100\ \Omega$, what is the range of R_X in Ohms?



- (A) [123.50, 136.50]
- (B) [125.89, 134.12]
- (C) [117.00, 143.00]
- (D) [120.25, 139.75]

Options :

1. ✓ A
2. ✗ B
3. ✗ C
4. ✗ D

Question Number : 16 Question Type : NAT

A (0-50 A) moving coil ammeter has a voltage drop of 0.1 V across its terminals at full scale deflection. The external shunt resistance (in milliohms) needed to extend its range to (0 – 500 A) is _____.

Correct Answer :

0.22 to 0.23

Question Number : 17 Question Type : MCQ

Of the four characteristics given below, which are the major requirements for an instrumentation amplifier?

- P. High common mode rejection ratio
- Q. High input impedance
- R. High linearity
- S. High output impedance

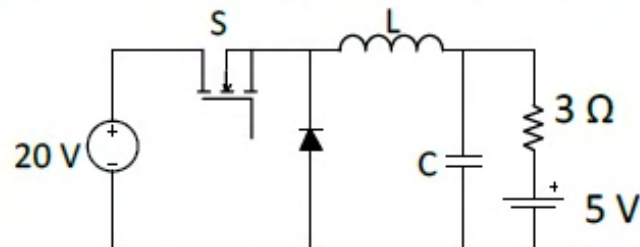
- | | |
|---------------------|---------------------|
| (A) P, Q and R only | (B) P and R only |
| (C) P, Q and S only | (D) Q, R and S only |

Options :

1. ✓ A
2. ✗ B
3. ✗ C
4. ✗ D

Question Number : 18 Question Type : NAT

In the following chopper, the duty ratio of switch S is 0.4. If the inductor and capacitor are sufficiently large to ensure continuous inductor current and ripple free capacitor voltage, the charging current (in Ampere) of the 5 V battery, under steady-state, is _____.



Correct Answer :

1

Question Number : 19 Question Type : NAT

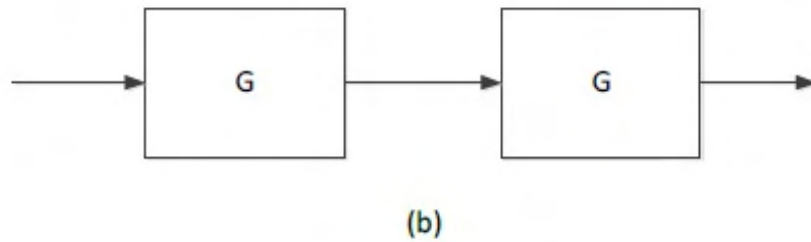
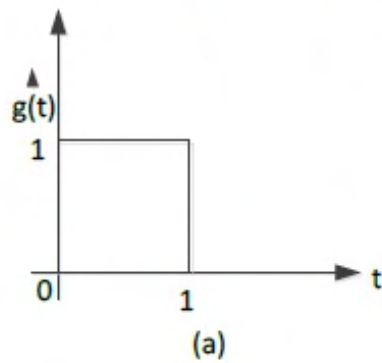
A moving average function is given by $y(t) = \frac{1}{T} \int_{t-T}^t u(\tau) d\tau$. If the input u is a sinusoidal signal of frequency $\frac{1}{2T}$ Hz, then in steady state, the output y will lag u (in degree) by _____.

Correct Answer :

90

Question Number : 20 Question Type : MCQ

The impulse response $g(t)$ of a system, G , is as shown in Figure (a). What is the maximum value attained by the impulse response of two cascaded blocks of G as shown in Figure (b)?



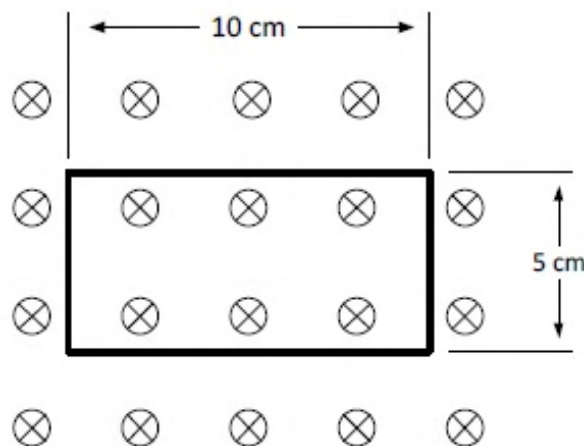
- (A) $\frac{2}{3}$ (B) $\frac{3}{4}$ (C) $\frac{4}{5}$ (D) 1

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 21 Question Type : NAT

Consider a one-turn rectangular loop of wire placed in a uniform magnetic field as shown in the figure. The plane of the loop is perpendicular to the field lines. The resistance of the loop is 0.4Ω , and its inductance is negligible. The magnetic flux density (in Tesla) is a function of time, and is given by $B(t) = 0.25 \sin \omega t$, where $\omega = 2\pi \times 50$ radian/second. The power absorbed (in Watt) by the loop from the magnetic field is _____.

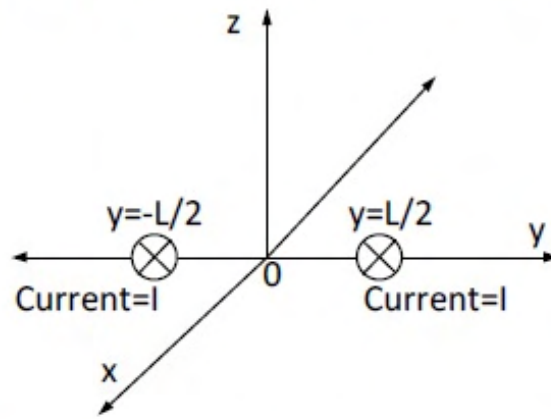


Correct Answer :

0.17 to 0.2

Question Number : 22 Question Type : MCQ

A steady current I is flowing in the $-x$ direction through each of two infinitely long wires at $y = \pm \frac{L}{2}$ as shown in the figure. The permeability of the medium is μ_0 . The $\vec{B}$ -field at $(0, L, 0)$ is



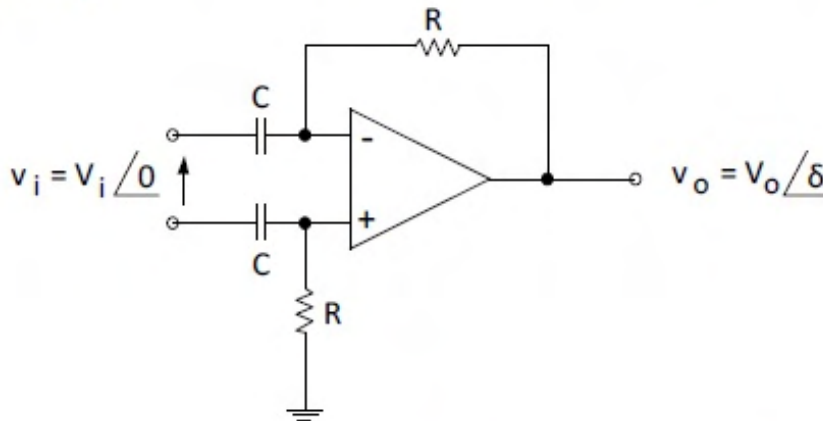
- (A) $-\frac{4\mu_0 I}{3\pi L} \hat{z}$ (B) $+\frac{4\mu_0 I}{3\pi L} \hat{z}$ (C) 0 (D) $-\frac{3\mu_0 I}{4\pi L} \hat{z}$

Options :

1. ✓ A
2. ✗ B
3. ✗ C
4. ✗ D

Question Number : 23 Question Type : MCQ

Consider the circuit shown in the figure. In this circuit $R=1 \text{ k}\Omega$, and $C=1 \text{ }\mu\text{F}$. The input voltage is sinusoidal with a frequency of 50 Hz, represented as a phasor with magnitude V_i and phase angle 0 radian as shown in the figure. The output voltage is represented as a phasor with magnitude V_o and phase angle δ radian. What is the value of the output phase angle δ (in radian) relative to the phase angle of the input voltage?



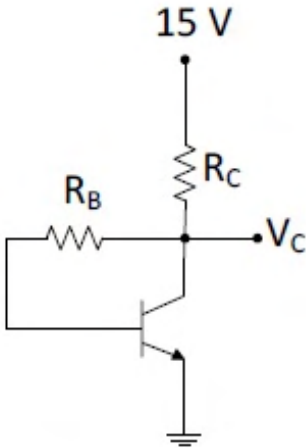
- (A) 0 (B) π (C) $\pi/2$ (D) $-\pi/2$

Options :

1. ✗ A
2. ✗ B
3. ✗ C
4. ✓ D

Question Number : 24 Question Type : NAT

In the given circuit, the silicon transistor has $\beta = 75$ and a collector voltage $V_C = 9\text{ V}$. Then the ratio of R_B and R_C is _____.

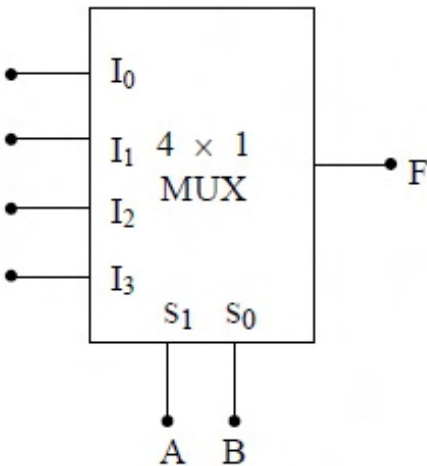


Correct Answer :

100 to 110

Question Number : 25 Question Type : MCQ

In the 4×1 multiplexer, the output F is given by $F = A \oplus B$. Find the required input ' $I_3I_2I_1I_0$ '.



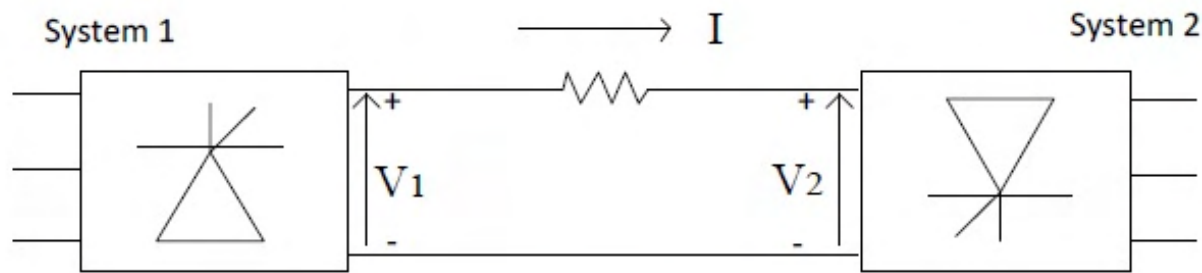
- (A) 1010
- (B) 0110
- (C) 1000
- (D) 1110

Options :

1.  A
2.  B
3.  C
4.  D

Question Number : 26 Question Type : MCQ

Consider a HVDC link which uses thyristor based line-commutated converters as shown in the figure. For a power flow of 750 MW from System 1 to System 2, the voltages at the two ends, and the current, are given by: $V_1 = 500$ kV, $V_2 = 485$ kV and $I = 1.5$ kA. If the direction of power flow is to be reversed (that is, from System 2 to System 1) without changing the electrical connections, then which one of the following combinations is feasible?



- (A) $V_1 = -500$ kV, $V_2 = -485$ kV and $I = 1.5$ kA
 (B) $V_1 = -485$ kV, $V_2 = -500$ kV and $I = 1.5$ kA
 (C) $V_1 = 500$ kV, $V_2 = 485$ kV and $I = -1.5$ kA
 (D) $V_1 = -500$ kV, $V_2 = -485$ kV and $I = -1.5$ kA

Options :

1. ✗ A
 2. ✓ B
 3. ✗ C
 4. ✗ D

Question Number : 27 Question Type : MCQ

Base load power plants are

- P: wind farms.
 Q: run-of-river plants.
 R: nuclear power plants.
 S: diesel power plants.

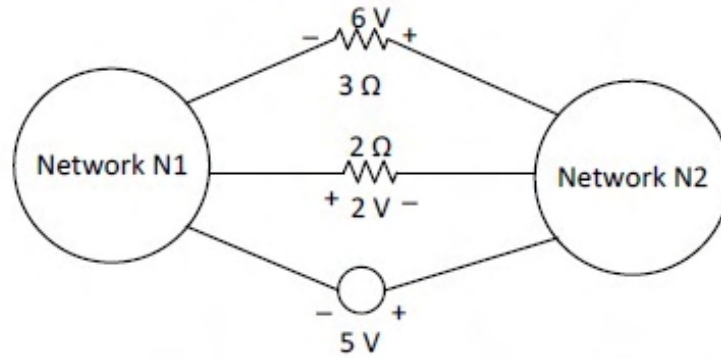
- (A) P, Q and S only
 (B) P, R and S only
 (C) P, Q and R only
 (D) Q and R only

Options :

1. ✗ A
 2. ✗ B
 3. ✓ C
 4. ✗ D

Question Number : 28 Question Type : MCQ

The voltages developed across the $3\ \Omega$ and $2\ \Omega$ resistors shown in the figure are 6 V and 2 V respectively, with the polarity as marked. What is the power (in Watt) delivered by the 5 V voltage source?



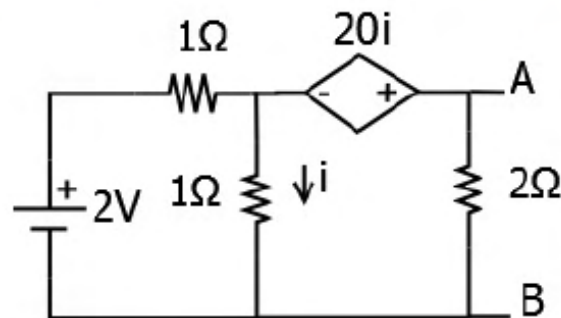
- (A) 5 (B) 7 (C) 10 (D) 14

Options :

1. ✓ A
2. ✗ B
3. ✗ C
4. ✗ D

Question Number : 29 Question Type : NAT

For the given circuit, the Thevenin equivalent is to be determined. The Thevenin voltage, V_{Th} (in Volt), seen from terminal AB is _____.

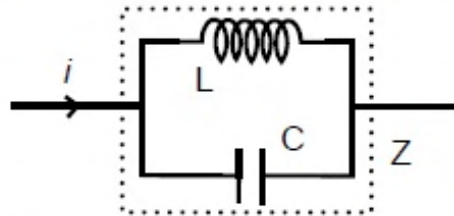


Correct Answer :

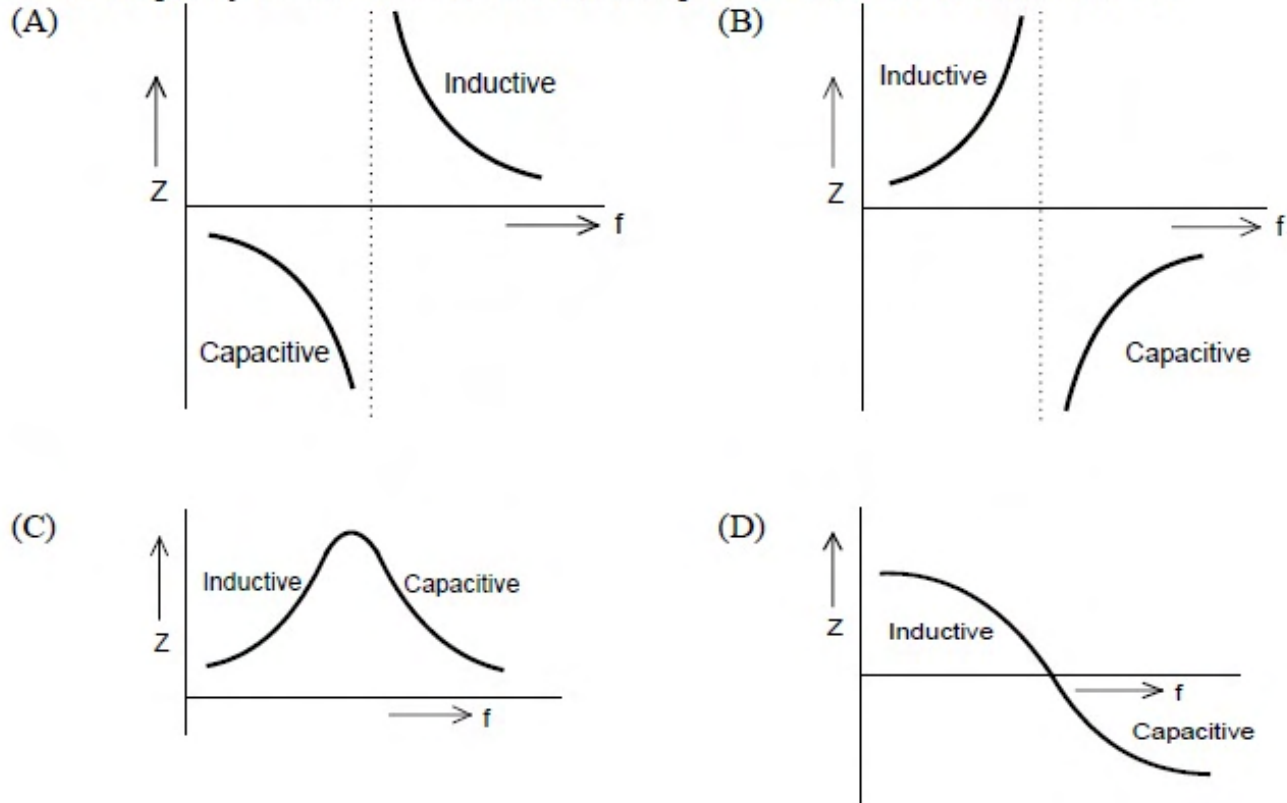
3.3 to 3.4

Question Number : 30 Question Type : MCQ

An inductor is connected in parallel with a capacitor as shown in the figure.



As the frequency of current i is increased, the impedance (Z) of the network varies as



Options :

1. ✗ A
2. ✓ B
3. ✗ C
4. ✗ D

Question Number : 31 Question Type : MCQ

A separately excited DC generator has an armature resistance of 0.1Ω and negligible armature inductance. At rated field current and rated rotor speed, its open-circuit voltage is 200 V. When this generator is operated at half the rated speed, with half the rated field current, an uncharged $1000\ \mu\text{F}$ capacitor is suddenly connected across the armature terminals. Assume that the speed remains unchanged during the transient. At what time (in microsecond) after the capacitor is connected will the voltage across it reach 25 V?

- (A) 62.25 (B) 69.3 (C) 73.25 (D) 77.3

Options :

1. ✗ A
2. ✓ B
3. ✗ C
4. ✗ D

Question Number : 32 Question Type : MCQ

The self inductance of the primary winding of a single phase, 50 Hz, transformer is 800 mH, and that of the secondary winding is 600 mH. The mutual inductance between these two windings is 480 mH. The secondary winding of this transformer is short circuited and the primary winding is connected to a 50 Hz, single phase, sinusoidal voltage source. The current flowing in both the windings is less than their respective rated currents. The resistance of both windings can be neglected. In this condition, what is the effective inductance (in mH) seen by the source?

- (A) 416 (B) 440 (C) 200 (D) 920

Options :

1. ✓ A
2. ✗ B
3. ✗ C
4. ✗ D

Question Number : 33 Question Type : MCQ

The primary mmf is least affected by the secondary terminal conditions in a

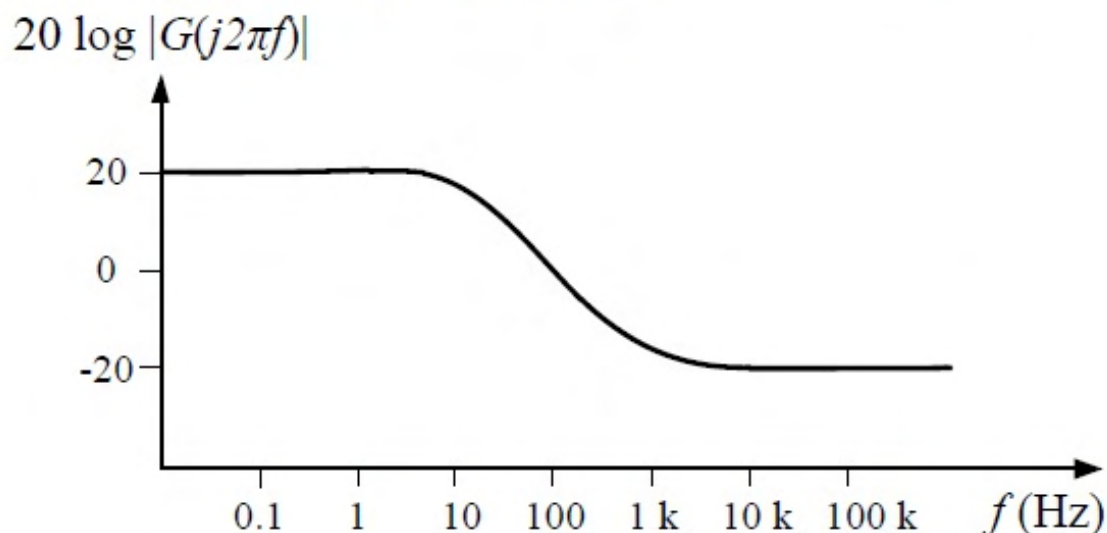
- (A) power transformer. (B) potential transformer.
(C) current transformer. (D) distribution transformer.

Options :

1. ✗ A
2. ✗ B
3. ✓ C
4. ✗ D

Question Number : 34 Question Type : MCQ

A Bode magnitude plot for the transfer function $G(s)$ of a plant is shown in the figure. Which one of the following transfer functions best describes the plant?



- (A) $\frac{1000(s+10)}{s+1000}$ (B) $\frac{10(s+10)}{s(s+1000)}$ (C) $\frac{s+1000}{10s(s+10)}$ (D) $\frac{s+1000}{10(s+10)}$

Options :

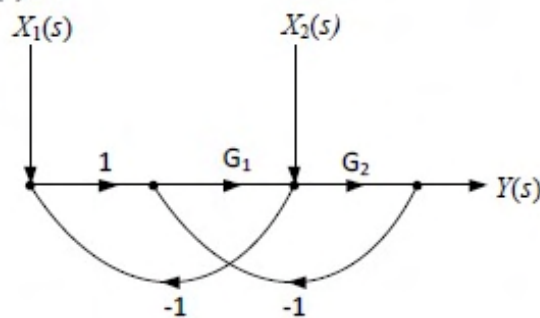
1. ✗ A
2. ✗ B

3. ✖ C

4. ✔ D

Question Number : 35 Question Type : MCQ

For the signal-flow graph shown in the figure, which one of the following expressions is equal to the transfer function $\left. \frac{Y(s)}{X_2(s)} \right|_{X_1(s)=0}$?



(A) $\frac{G_1}{1+G_2(1+G_1)}$

(B) $\frac{G_2}{1+G_1(1+G_2)}$

(C) $\frac{G_1}{1+G_1G_2}$

(D) $\frac{G_2}{1+G_1G_2}$

Options :

1. ✖ A

2. ✔ B

3. ✖ C

4. ✖ D

Question Number : 36 Question Type : MCQ

The maximum value of “a” such that the matrix $\begin{pmatrix} -3 & 0 & -2 \\ 1 & -1 & 0 \\ 0 & a & -2 \end{pmatrix}$ has three linearly independent real eigenvectors is

(A) $\frac{2}{3\sqrt{3}}$

(B) $\frac{1}{3\sqrt{3}}$

(C) $\frac{1+2\sqrt{3}}{3\sqrt{3}}$

(D) $\frac{1+\sqrt{3}}{3\sqrt{3}}$

Options :

1. ✖ A

2. ✔ B

3. ✖ C

4. ✖ D

Question Number : 37 Question Type : NAT

A solution of the ordinary differential equation $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0$ is such that $y(0) = 2$ and $y(1) = -\frac{1-3e}{e^3}$. The value of $\frac{dy}{dt}(0)$ is _____.

Correct Answer :

-3

Question Number : 38 Question Type : MCQ

The signum function is given by

$$\operatorname{sgn}(x) = \begin{cases} \frac{x}{|x|}; & x \neq 0 \\ 0; & x = 0 \end{cases}$$

The Fourier series expansion of $\operatorname{sgn}(\cos(t))$ has

- (A) only sine terms with all harmonics.
- (B) only cosine terms with all harmonics.
- (C) only sine terms with even numbered harmonics.
- (D) only cosine terms with odd numbered harmonics.

Options :

- 1. ✖ A
- 2. ✖ B
- 3. ✖ C
- 4. ✔ D

Question Number : 39 Question Type : MCQ

Two players, A and B, alternately keep rolling a fair dice. The person to get a six first wins the game. Given that player A starts the game, the probability that A wins the game is

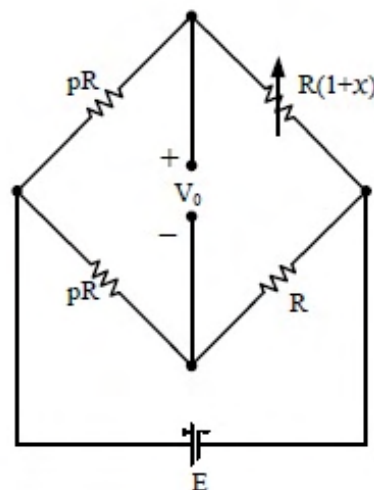
- (A) $5/11$ (B) $1/2$ (C) $7/13$ (D) $6/11$

Options :

- 1. ✖ A
- 2. ✖ B
- 3. ✖ C
- 4. ✔ D

Question Number : 40 Question Type : MCQ

An unbalanced DC Wheatstone bridge is shown in the figure. At what value of p will the magnitude of V_0 be maximum?



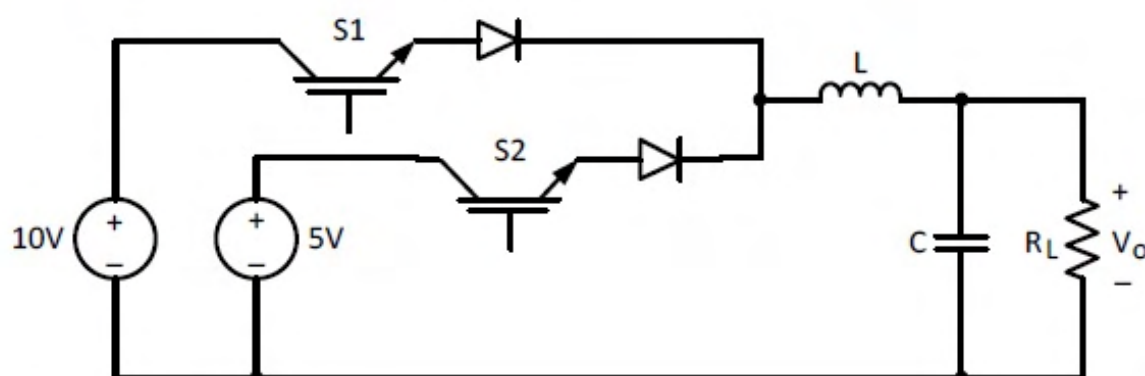
- (A) $\sqrt{(1+x)}$ (B) $(1+x)$ (C) $1/\sqrt{(1+x)}$ (D) $\sqrt{(1-x)}$

Options :

1. ✓ A
2. ✗ B
3. ✗ C
4. ✗ D

Question Number : 41 Question Type : NAT

The circuit shown is meant to supply a resistive load R_L from two separate DC voltage sources. The switches S1 and S2 are controlled so that only one of them is ON at any instant. S1 is turned on for 0.2 ms and S2 is turned on for 0.3 ms in a 0.5 ms switching cycle time period. Assuming continuous conduction of the inductor current and negligible ripple on the capacitor voltage, the output voltage V_O (in Volt) across R_L is _____.

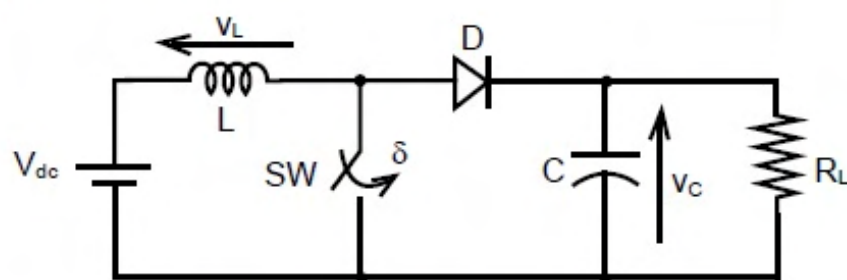


Correct Answer :

7

Question Number : 42 Question Type : MCQ

A self commutating switch SW, operated at duty cycle δ is used to control the load voltage as shown in the figure



Under steady state operating conditions, the average voltage across the inductor and the capacitor respectively, are

(A) $V_L = 0$ and $V_C = \frac{1}{1-\delta} V_{dc}$

(B) $V_L = \frac{\delta}{2} V_{dc}$ and $V_C = \frac{1}{1-\delta} V_{dc}$

(C) $V_L = 0$ and $V_C = \frac{\delta}{1-\delta} V_{dc}$

(D) $V_L = \frac{\delta}{2} V_{dc}$ and $V_C = \frac{\delta}{1-\delta} V_{dc}$

Options :

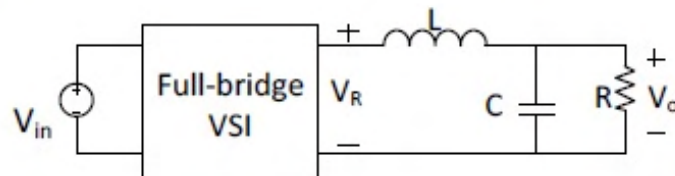
1. ✓ A
2. ✗ B

3. ✖ C

4. ✖ D

Question Number : 43 Question Type : NAT

The single-phase full-bridge voltage source inverter (VSI), shown in figure, has an output frequency of 50 Hz. It uses unipolar pulse width modulation with switching frequency of 50 kHz and modulation index of 0.7. For $V_{in} = 100$ V DC, $L = 9.55$ mH, $C = 63.66$ μ F, and $R = 5$ Ω , the amplitude of the fundamental component in the output voltage V_o (in Volt) under steady-state is _____.



Correct Answer :

60 to 64

Question Number : 44 Question Type : NAT

A 3-phase 50 Hz square wave (6-step) VSI feeds a 3-phase, 4 pole induction motor. The VSI line voltage has a dominant 5th harmonic component. If the operating slip of the motor with respect to fundamental component voltage is 0.04, the slip of the motor with respect to 5th harmonic component of voltage is _____.

Correct Answer :

1.16 to 1.22

Question Number : 45 Question Type : MCQ

Consider a discrete time signal given by

$$x[n] = (-0.25)^n u[n] + (0.5)^n u[-n - 1]$$

The region of convergence of its Z-transform would be

- (A) the region inside the circle of radius 0.5 and centered at origin
- (B) the region outside the circle of radius 0.25 and centered at origin
- (C) the annular region between the two circles, both centered at origin and having radii 0.25 and 0.5
- (D) the entire Z plane.

Options :

1. ✖ A

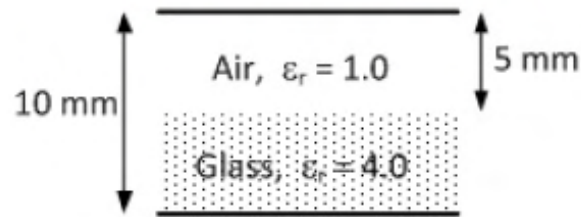
2. ✖ B

3. ✔ C

4. ✖ D

Question Number : 46 Question Type : NAT

A parallel plate capacitor is partially filled with glass of dielectric constant 4.0 as shown below. The dielectric strengths of air and glass are 30 kV/cm and 300 kV/cm, respectively. The maximum voltage (in kilovolts), which can be applied across the capacitor without any breakdown, is _____.

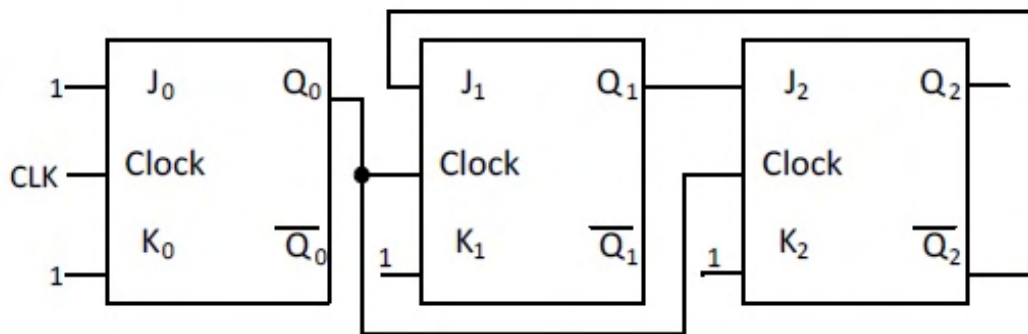


Correct Answer :

17 to 20

Question Number : 47 Question Type : NAT

The figure shows a digital circuit constructed using negative edge triggered J-K flip flops. Assume a starting state of $Q_2Q_1Q_0=000$. This state $Q_2Q_1Q_0=000$ will repeat after _____ number of cycles of the clock CLK.



Correct Answer:

6

Question Number : 48 Question Type : MCQ

$f(A, B, C, D) = \Pi M(0, 1, 3, 4, 5, 7, 9, 11, 12, 13, 14, 15)$ is a maxterm representation of a Boolean function $f(A, B, C, D)$ where A is the MSB and D is the LSB. The equivalent minimized representation of this function is

- (A) $(A + \overline{C} + D)(\overline{A} + B + D)$
- (B) $A\overline{C}D + \overline{A}BD$
- (C) $\overline{A}\overline{C}\overline{D} + A\overline{B}C\overline{D} + A\overline{B}\overline{C}\overline{D}$
- (D) $(B + \overline{C} + D)(A + \overline{B} + \overline{C} + D)(\overline{A} + B + C + D)$

Options :

1. ✓ A

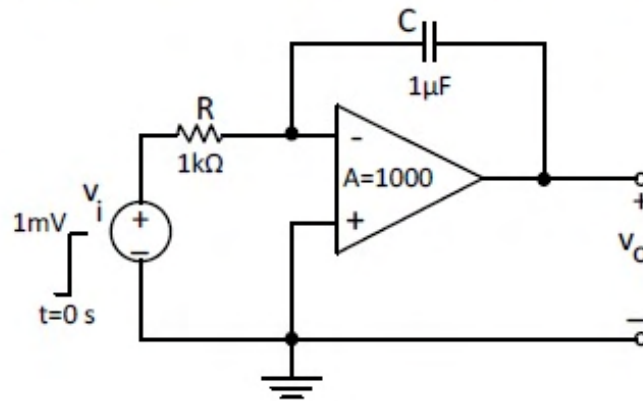
2. ✗ B

3. ✗ C

4. ✗ D

Question Number : 49 Question Type : MCQ

The op-amp shown in the figure has a finite gain $A = 1000$ and an infinite input resistance. A step-voltage $V_i = 1 \text{ mV}$ is applied at the input at time $t = 0$ as shown. Assuming that the operational amplifier is not saturated, the time constant (in millisecond) of the output voltage V_o is



- (A) 1001
- (B) 101
- (C) 11
- (D) 1

Options :

- 1. ✓ A
- 2. ✗ B
- 3. ✗ C
- 4. ✗ D

Question Number : 50 Question Type : MCQ

An 8-bit, unipolar Successive Approximation Register type ADC is used to convert 3.5 V to digital equivalent output. The reference voltage is +5 V. The output of the ADC, at the end of 3rd clock pulse after the start of conversion, is

- | | |
|---------------|---------------|
| (A) 1010 0000 | (B) 1000 0000 |
| (C) 0000 0001 | (D) 0000 0011 |

Options :

- 1. ✓ A
- 2. ✗ B
- 3. ✗ C
- 4. ✗ D

Question Number : 51 Question Type : NAT

Consider the economic dispatch problem for a power plant having two generating units. The fuel costs in Rs/MWh along with the generation limits for the two units are given below:

$$C_1(P_1) = 0.01P_1^2 + 30P_1 + 10 \quad ; \quad 100 \text{ MW} \leq P_1 \leq 150 \text{ MW}$$

$$C_2(P_2) = 0.05P_2^2 + 10P_2 + 10 \quad ; \quad 100 \text{ MW} \leq P_2 \leq 180 \text{ MW}$$

The incremental cost (in Rs/MWh) of the power plant when it supplies 200 MW is _____ .

Correct Answer :

20

Question Number : 52 Question Type : MCQ

Determine the correctness or otherwise of the following Assertion [a] and the Reason [r].

Assertion: Fast decoupled load flow method gives approximate load flow solution because it uses several assumptions.

Reason: Accuracy depends on the power mismatch vector tolerance.

- (A) Both [a] and [r] are true and [r] is the correct reason for [a].
- (B) Both [a] and [r] are true but [r] is not the correct reason for [a].
- (C) Both [a] and [r] are false.
- (D) [a] is false and [r] is true.

Options :

- 1. ✖ A
- 2. ✖ B
- 3. ✖ C
- 4. ✔ D

Question Number : 53 Question Type : NAT

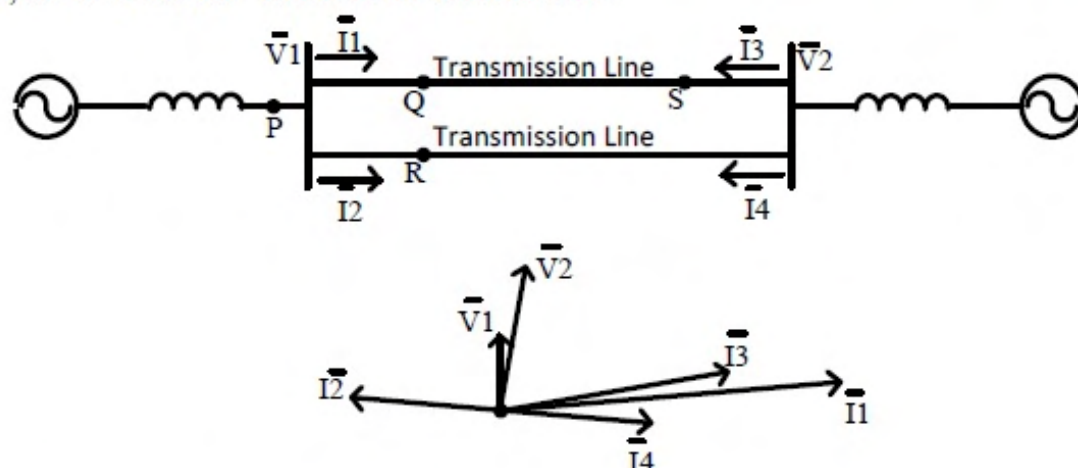
A 50 Hz generating unit has H -constant of 2 MJ/MVA. The machine is initially operating in steady state at synchronous speed, and producing 1 pu of real power. The initial value of the rotor angle δ is 5° , when a bolted three phase to ground short circuit fault occurs at the terminal of the generator. Assuming the input mechanical power to remain at 1 pu, the value of δ in degrees, 0.02 second after the fault is _____.

Correct Answer :

5.7 to 6.1

Question Number : 54 Question Type : MCQ

A sustained three-phase fault occurs in the power system shown in the figure. The current and voltage phasors during the fault (on a common reference), after the natural transients have died down, are also shown. Where is the fault located?



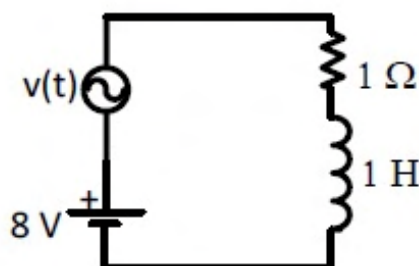
- (A) Location P (B) Location Q (C) Location R (D) Location S

Options :

1. ✗ A
2. ✓ B
3. ✗ C
4. ✗ D

Question Number : 55 Question Type : NAT

The circuit shown in the figure has two sources connected in series. The instantaneous voltage of the AC source (in Volt) is given by $v(t) = 12 \sin t$. If the circuit is in steady state, then the rms value of the current (in Ampere) flowing in the circuit is _____.



Correct Answer :

9.9 to 10.1

Question Number : 56 Question Type : NAT

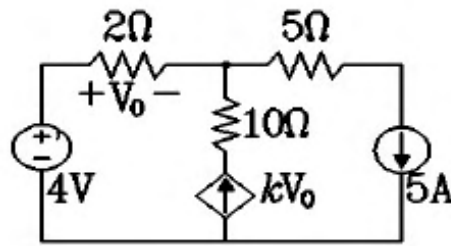
In a linear two-port network, when 10 V is applied to Port 1, a current of 4 A flows through Port 2 when it is short-circuited. When 5 V is applied to Port 1, a current of 1.25 A flows through a 1Ω resistance connected across Port 2. When 3 V is applied to Port 1, the current (in Ampere) through a 2Ω resistance connected across Port 2 is _____.

Correct Answer:

0.4 to 0.6

Question Number : 57 Question Type : NAT

In the given circuit, the parameter k is positive, and the power dissipated in the $2\ \Omega$ resistor is 12.5W . The value of k is _____.



Correct Answer :

0.48 to 0.52

Question Number : 58 Question Type : NAT

A separately excited DC motor runs at 1000 rpm on no load when its armature terminals are connected to a 200V DC source and the rated voltage is applied to the field winding. The armature resistance of this motor is $1\ \Omega$. The no-load armature current is negligible. With the motor developing its full load torque, the armature voltage is set so that the rotor speed is 500 rpm. When the load torque is reduced to 50% of the full load value under the same armature voltage conditions, the speed rises to 520 rpm. Neglecting the rotational losses, the full load armature current (in Ampere) is _____.

Correct Answer:

8

Question Number : 59 Question Type : NAT

A DC motor has the following specifications: 10 hp, 37.5 A, 230 V; flux/pole = 0.01 Wb, number of poles = 4, number of conductors = 666, number of parallel paths = 2. Armature resistance = $0.267\ \Omega$. The armature reaction is negligible and rotational losses are 600 W. The motor operates from a 230 V DC supply. If the motor runs at 1000 rpm, the output torque produced (in Nm) is _____.

Correct Answer :

57 to 58

Question Number : 60 Question Type : NAT

A 200/400 V, 50 Hz, two-winding transformer is rated at 20 kVA. Its windings are connected as an auto-transformer of rating 200/600 V. A resistive load of $12\ \Omega$ is connected to the high voltage (600 V) side of the auto-transformer. The value of equivalent load resistance (in Ohm) as seen from low voltage side is _____.

Correct Answer :

1.3 to 1.4

Question Number : 61 Question Type : NAT

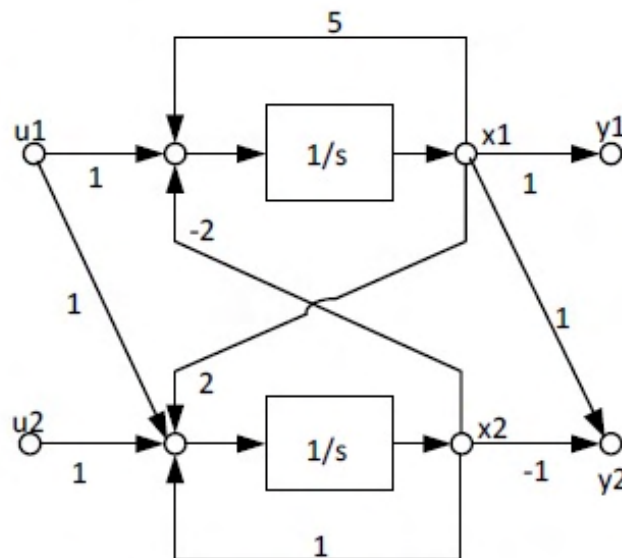
Two single-phase transformers T_1 and T_2 each rated at 500 kVA are operated in parallel. Percentage impedances of T_1 and T_2 are $(1 + j6)$ and $(0.8 + j4.8)$, respectively. To share a load of 1000 kVA at 0.8 lagging power factor, the contribution of T_2 (in kVA) is _____.

Correct Answer :

554 to 556

Question Number : 62 Question Type : MCQ

In the signal flow diagram given in the figure, u_1 and u_2 are possible inputs whereas y_1 and y_2 are possible outputs. When would the SISO system derived from this diagram be controllable and observable?



- (A) When u_1 is the only input and y_1 is the only output.
- (B) When u_2 is the only input and y_1 is the only output.
- (C) When u_1 is the only input and y_2 is the only output.
- (D) When u_2 is the only input and y_2 is the only output.

Options :

- 1. ✗ A
- 2. ✓ B

3. ✖ C

4. ✖ D

Question Number : 63 Question Type : MCQ

The transfer function of a second order real system with a perfectly flat magnitude response of unity has a pole at $(2 - j3)$. List all the poles and zeroes.

(A) Poles at $(2 \pm j3)$, no zeroes.

(B) Poles at $(\pm 2 - j3)$, one zero at origin.

(C) Poles at $(2 - j3)$, $(-2 + j3)$, zeroes at $(-2 - j3)$, $(2 + j3)$.

(D) Poles at $(2 \pm j3)$, zeroes at $(-2 \pm j3)$.

Options :

1. ✖ A

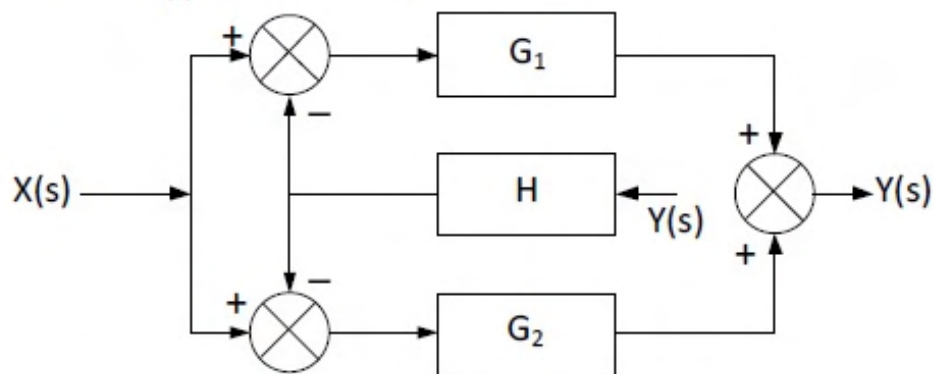
2. ✖ B

3. ✖ C

4. ✔ D

Question Number : 64 Question Type : MCQ

Find the transfer function $\frac{Y(s)}{X(s)}$ of the system given below.



(A) $\frac{G_1}{1-HG_1} + \frac{G_2}{1-HG_2}$

(B) $\frac{G_1}{1+HG_1} + \frac{G_2}{1+HG_2}$

(C) $\frac{G_1+G_2}{1+H(G_1+G_2)}$

(D) $\frac{G_1+G_2}{1-H(G_1+G_2)}$

Options :

1. ✖ A

2. ✖ B

3. ✔ C

4. ✖ D

Question Number : 65 Question Type : MCQ

The open loop poles of a third order unity feedback system are at $0, -1, -2$. Let the frequency corresponding to the point where the root locus of the system transits to unstable region be K . Now suppose we introduce a zero in the open loop transfer function at -3 , while keeping all the earlier open loop poles intact. Which one of the following is TRUE about the point where the root locus of the modified system transits to unstable region?

- (A) It corresponds to a frequency greater than K
- (B) It corresponds to a frequency less than K
- (C) It corresponds to a frequency K
- (D) Root locus of modified system never transits to unstable region

Options :

- 1. ✖ A
- 2. ✖ B
- 3. ✖ C
- 4. ✔ D